

TUTOR-60

Does buying fewer tutoring minutes for every eligible student beat buying the pilot's ninety weekly minutes for the fraction of them the budget reaches?

Abstract

This study examined the allocation of tutoring minutes under a fixed budget for eligible students. The question asked was whether buying fewer tutoring minutes for every eligible student beats buying the pilot allocation's ninety weekly minutes for the fraction of them the budget reaches. This was answered by comparing cohort reading points over the pilot allocation against a decision threshold of zero points per year. The effect of tutoring on gain is identified via the backdoor route in the randomization design. The analysis found that cohort reading points over the pilot allocation settled above the threshold on the favourable side at 13838 points per year with a 90% HDI from 6163 to 22036 points per year. The value of the difference at the department's own valuation also settled above the threshold on the favourable side. These findings depend on the unverified assumption that the cohort average answers the question. The results indicate that spreading a lower dose across all eligible students delivers more total cohort reading points than funding a high dose for a smaller fraction of the students.

Reported quantities: Cohort reading points over the pilot allocation; Value of the difference at the department's own valuation

1. Introduction

This report addresses whether buying fewer tutoring minutes for every eligible student beats buying the pilot's ninety weekly minutes for the fraction of them the budget reaches. The target quantity is identified. The question is answered by cohort reading points over the pilot allocation, evaluated against a decision threshold of 0 points/year. An interval that lies entirely on one side of 0 points/year settles the question. An interval strictly above 0 points/year indicates that allocating fewer minutes to every eligible student yields higher cohort reading points than the pilot allocation. An interval strictly below 0 points/year indicates that the pilot allocation yields higher cohort reading points. An interval spanning 0 points/year leaves the comparison unsettled.

2. Methods

To address the question introduced earlier, the target is the effect of tutoring on gain, identified via the backdoor route. The graph consists of poverty_share → gain, resources → gain, resources → poverty_share, and tutoring → gain, where resources are unmeasured, as shown in Table 1 and Figure 1. The trial design parameters are shown in Table 3. Across that design, 120 schools were randomized to allocations drawn from a lattice over the two doses, with 3 messaging levels and tutoring spread across the decision range, leaving the adjustment set empty. Relying on state records on 240 schools was considered instead, but in that graph poverty_share → gain, poverty_share → tutoring, resources → gain, resources → poverty_share, resources → tutoring, and tutoring → gain, which downgrades the effect because resources are unmeasured. A two-arm trial at the pilot's 90 minutes was also considered and rejected because it would only answer whether tutoring works rather than the shape of the response. Treatments are dosed in dollars per eligible student per year—where 20 USD buys a weekly tutoring minute and 9 USD buys a weekly caregiver message capped at 5—under a budget of 9,000,000 USD for one school year across 12,000 students reading two or more grade levels behind, equaling 750 USD a student if every one is reached. Dosing tutoring in minutes and messaging in messages was considered and rejected because the two

Planning the trial

```
assignment: (tutoring_s, messaging_s) ~ Uniform(lattice) per school s
observational graph : poverty_share → gain, poverty_share → tutoring,
resources → gain, resources → poverty_share, resources → tutoring,
tutoring → gain
status downgraded - requires the unmeasured node
randomized graph : poverty_share → gain, resources → gain, resources
→ poverty_share, tutoring → gain
status identified via backdoor
```

Estimating the response

```
gain_s = f(tutoring_s) + g(messaging_s) + eps_s
f, g = penalised splines, one per dose, on the school-level panel
effect(a) = E[gain | dose = a] - E[gain | dose = 0] within each draw
served(a) = min(N_eligible, Budget / cost(a))
cohort(a) = served(a) * effect(a) the objective
cost(a) = a_tutoring + a_messaging (USD per eligible student-year)
```

Choosing the allocation

```
a* = argmax_a mean_draws cohort(a)
Delta = cohort(a*) - cohort(a_pilot) paired inside each draw
P = Pr(Delta > 0)
value = Delta * VALUE_PER_POINT
```

The identification graph

From	Arrow	To	Kind
poverty_share	→	gain	direct effect
resources	→	gain	direct effect
resources	→	poverty_share	direct effect

tutoring	→	gain	direct effect
resources			unmeasured node

Table 1. The identification graph, as arrows

Assumptions this analysis rests on

Assumption	Statement	Status	Challenged by
a reading point has the value the state assigns it	a reading point is valued at 1,400 USD; the recommendation does not depend on this number, only on whether to fund at all	was asserted without a check	a valuation low enough to make the whole program not worth funding
the cohort average answers the question	the recommendation is made on the cohort average; the grade-band breakdown the program office asked for is not estimable from a school-randomized trial	is assumed and has not been checked	a response that differs across grade bands enough to change the allocation

Table 2. Standing assumptions

identification Schools were randomized to allocations; the adjustment set is empty.

assumes *schools_were_randomized_to_allocations* (satisfied): schools were randomized to allocations, so the adjustment set is empty

estimand The recommendation is over assigned allocations. Schools delivered 93% of assigned minutes, and that shortfall is inside the estimate rather than corrected out of it.

assumes *the_estimand_is_over_assigned_allocations* (satisfied): the recommendation is over assigned allocations; schools delivered 93% of assigned minutes and that shortfall is priced into the estimate

interval Every figure and quantity at 90% over 4,000 posterior draws.

limitation The grade-band breakdown the program office asked for is not estimable from a school-randomized trial, and is not reported.

assumes *the_cohort_average_answers_the_question* (unverified): the recommendation is made on the cohort average; the grade-band breakdown the program office asked for is not estimable from a school-randomized trial

Assumption ledger

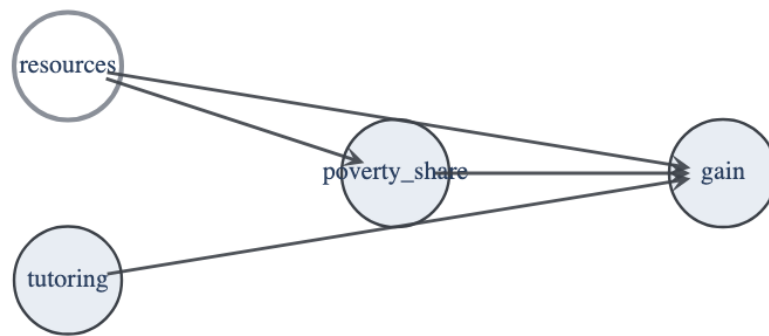


Figure 1. The identification graph. A hollow node was not measured and a dashed edge is an unmeasured common cause; those two are what decide whether the effect is identifiable at all

Step	Parameter	Value
Identification	graph	poverty_share -> gain, resources -> gain, resources -> poverty_share, tutoring -> gain
Identification	graph hash	488a01d83085
Identification	route	backdoor
Identification	status	identified
Identification	unmeasured nodes	resources
The decision as arithmetic	budget (USD)	9,000,000
The decision as arithmetic	eligible students	12,000
The decision as arithmetic	messaging dose range (USD)	0 to 45
The decision as arithmetic	per student if all are reached (USD)	750
The decision as arithmetic	tutoring dose range (USD)	0 to 3,000
Planning the trial	eligible per school	35–65
Planning the trial	messaging levels	0, 18, 45
Planning the trial	schools	120
Planning the trial	unit of assignment	school
Estimating the response	fit hash	c878161a12a5
Estimating the response	interval mass	90%

Estimating the response	posterior draws	4,000
Estimating the response	response family	spline in both doses
Estimating the response	sampler	PyMC via nutpie
Choosing the allocation	allocate() at the budget	tutoring 705, messaging 45 (converged)
Choosing the allocation	messages a week	5
Choosing the allocation	recommended messaging (USD)	45
Choosing the allocation	recommended tutoring (USD)	700
Choosing the allocation	students reached	12,000
Choosing the allocation	weekly minutes	35
Choosing the allocation	within 2% of the best (USD)	700 to 750

Table 3. The design, as recorded parameters

3. Results

The results address the comparison between the program allocation and the pilot allocation for the decision values established in the introduction. Following the design and model assumptions outlined in the methods, the analysis provides estimates for the primary outcomes. The cohort reading points over the pilot allocation is 13838 points/year (90% HDI 6163 to 22036 points/year). The value of the difference at the department's own valuation is 19372837 USD (90% HDI 8627873 to 30850017 USD). The estimated quantities are reported in Table 4. The estimated quantities with their intervals are displayed in Figure 2.

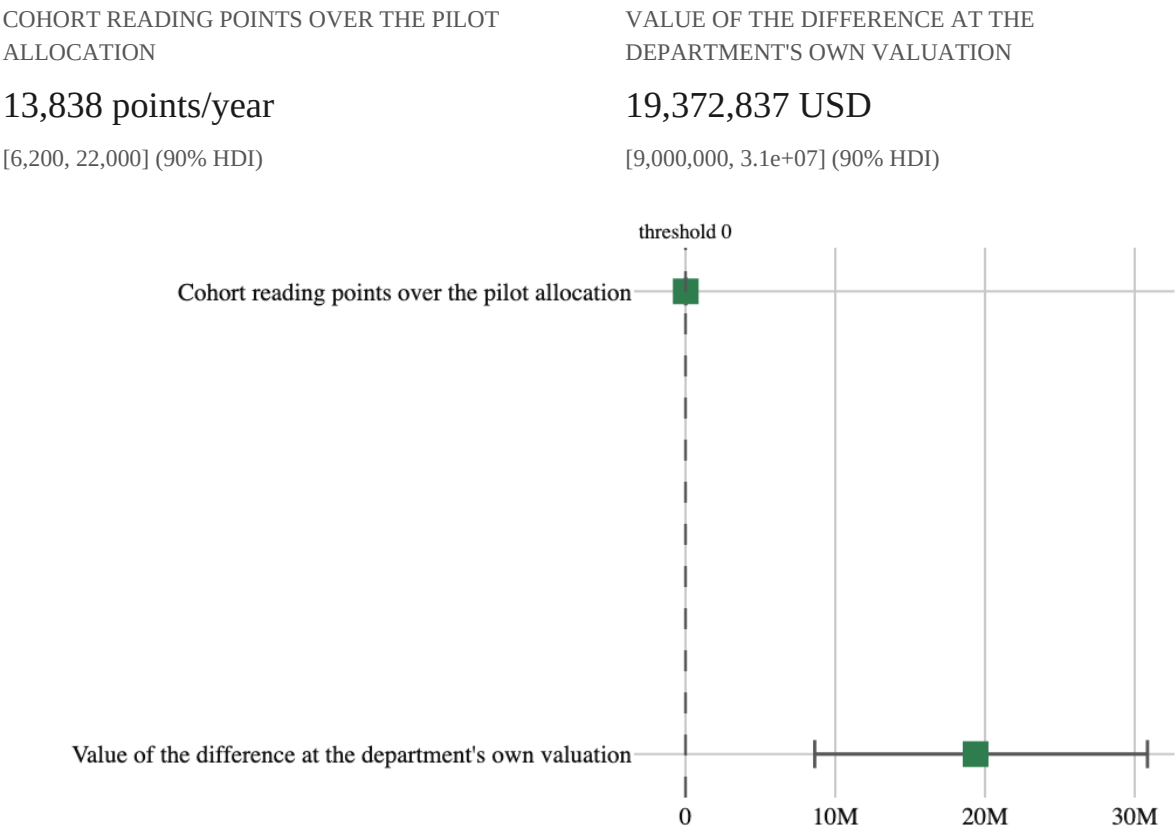


Figure 2. Estimated quantities with their intervals

Quantity	Estimate	Unit	Interval	Threshold	Relative to threshold
Cohort reading points over the pilot allocation	13838	points/year	6163 to 22036 (90% HDI)	0	above threshold
Value of the difference at the department's own valuation	19372837	USD	8627873 to 30850017 (90% HDI)	0	above threshold

Table 4. Estimated quantities

4. What the run showed

Written by the analyst after seeing the output, and reproduced unchanged. Nothing in this section is generated.

The recommendation is flat between 700 and 750 USD — 35 to 38 weekly minutes are all within two per cent of the best. The decision that matters is not the exact figure, it is not funding 90 minutes.

For the pilot to be the better buy, the advantage would have to be overstated by 100% of itself — that is, essentially the whole of it would have to be an artefact of the fit rather than a real difference between the two allocations.

The reason the recommendation is not 90 minutes is not that 90 minutes fails. It is that the response is steep at the bottom, the budget is fixed, and every extra dollar a student is a student not served.

5. Model checking

Diagnostic checks evaluate the fit and can lower confidence in an estimate, but they can never establish one. The estimation relies on 4000 posterior draws across the 120 randomized schools. The posterior probability that the recommendation beats the pilot is 0.997. Under the recommended allocation, the intervention reaches 12000 students. By contrast, the pilot allocation would reach 4878 students. The cost of the recommended allocation is 745 USD per student. Recorded diagnostics are presented in Table 5. This evaluation examines the sampling performance and the reach quantities generated by the model. It leaves unexamined the structural and identification conditions detailed in the methods. Diagnostic consistency reflects computational stability without verifying the broader foundational assumptions of the design.

POSTERIOR PROBABILITY THE RECOMMEN- DATION BEATS THE PILOT	STUDENTS THE RECOMMENDA- TION REACHES	STUDENTS THE PILOT ALLOCATION WOULD REACH	COST OF THE R ECOMMENDED ALLOCATION PER STUDENT	SCHOOLS RANDOMIZED	POSTERIOR DRAWS
0.997	12,000	4,878	745 USD	120	4,000

Check	Value	Unit	Interval
Posterior probability the recommendation beats the pilot	0.997	—	—
Students the recommendation reaches	12000	—	—
Students the pilot allocation would reach	4878	—	—
Cost of the recommended allocation per student	745	USD	—
Schools randomized	120	—	—
Posterior draws	4000	—	—

Table 5. Recorded diagnostics

6. Discussion

Both key findings agree and settle above the threshold on the favourable side. Specifically, the value of the difference at the department's own valuation and the cohort reading points over the pilot allocation each lie on the side favoring the broader distribution of tutoring minutes over the pilot allocation. Because the analysis relies on the backdoor route identified in the methods, these estimates can be read as effects of the allocation choice rather than baseline associations. That reading holds under the identification strategy but remains conditional on 2 unresolved assumptions. The limitations section states what each of those assumptions claims and what would challenge it.

7. Conclusions

Buying fewer tutoring minutes for every eligible student beats buying the pilot's ninety weekly minutes for the fraction of students reached by the budget. Cohort reading points over the pilot allocation settle on the favourable side of the decision threshold at 13838 points/year (90% HDI 6163 to 22036 points/year). The value of the difference at the department's own valuation also settles on the favourable side of the threshold. This finding follows from the identification set out earlier in the report. The conclusion is conditional on 2 unresolved assumptions: a reading point has the value the state assigns it, and the cohort average answers the question. What each assumption claims is stated in the limitations section. The policy answer holds under these conditions.

8. Limitations

Two unresolved assumptions support the analysis beyond what the data supply. First, the analysis asserts that a reading point has the value the state assigns it. Under this claim, a reading point is valued at 1,400 USD. This licenses the recommendation because the decision does not depend on this number, only on whether to fund at all. The assertion would be challenged by a valuation low enough to make the whole program not worth funding. Second, the analysis leaves unverified the assumption that the cohort average answers the question. This licenses basing the recommendation on the cohort average, as the grade-band breakdown the program office asked for is not estimable from a school-randomized trial. This assumption would be challenged by a response that differs across grade bands enough to change the allocation.

9. Provenance

Every number in this document resolves to a quantity in the evidence record below. Prose was checked against that record: any numeral not traceable to it, and any claim the record does not itself make, is reported rather than published.

Quantity	Key	Value	Produced by
Cohort reading points over the pilot allocation	advantage	13838 points/year (90% HDI 6163 to 22036 points/year)	paired posterior comparison over the allocation grid
Value of the difference at the department's own valuation	advantage_value	19372837 USD (90% HDI 8627873 to 30850017 USD)	the same draws, priced
Posterior probability the recommendation beats the pilot	probability_better	0.997	—
Students the recommendation reaches	students_reached	12000	—
Students the pilot allocation would reach	pilot_reached	4878	—
Cost of the recommended allocation per student	cost_per_student	745 USD	—
Schools randomized	schools	120	—
Posterior draws	draws	4000	—

Table 6. Quantities and their sources

Field	Value
estimator	spline response surface, PyMC via nutpie
fit_hash	c878161a12a5
interval_mass	90%
seed	0
world	nbs/case-studies/tutoring/tutoring.py
evidence hash	7bc9d4b50db1c7e7d179ec9a8af11003732629bc0189790cf210b75d82f12a52
narration: abstract	gemini-3.5-flash-lite — verified
narration: introduction	gemini-3.7-flash — verified
narration: methods	gemini-3.7-flash — verified
narration: results	gemini-3.7-flash — verified
narration: diagnostics	gemini-3.7-flash — verified
narration: discussion	gemini-3.7-flash — verified
narration: conclusions	gemini-3.7-flash — verified

Table 7. Run